The Future of (Endurance) Training

Reflections on some old and new coaching challenges and opportunities

Stephen Seiler PhD
University of Agder
Kristiansand, Norway

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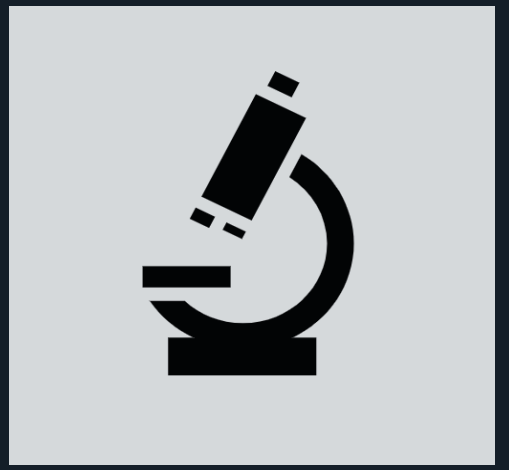
Sports Science- Dual directionality

We use science to improve sports performance



We study human function through the lens of sports training and performance

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What single technological innovation has had the **most profound impact** on **sport** and the **sports training** process?

1816

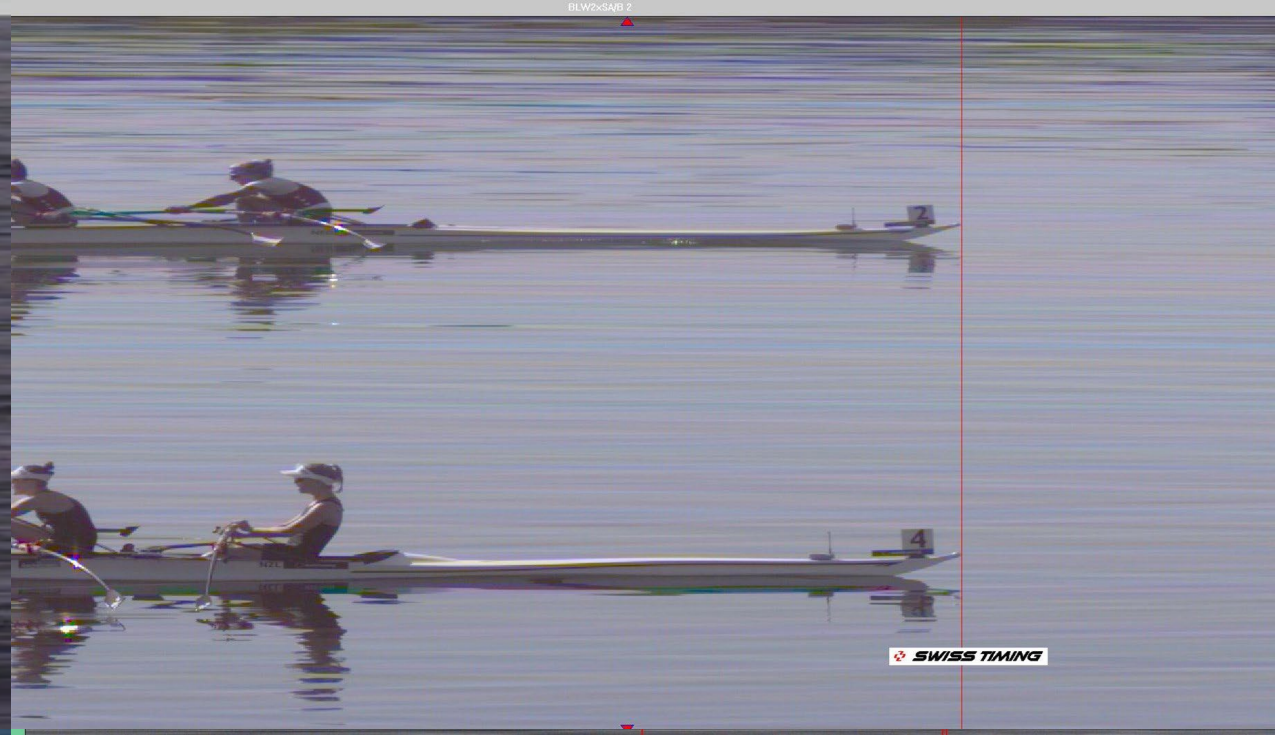
1932 Olympics

30 Stopwatches supplied by Omega to cover 14 sports and 117 events

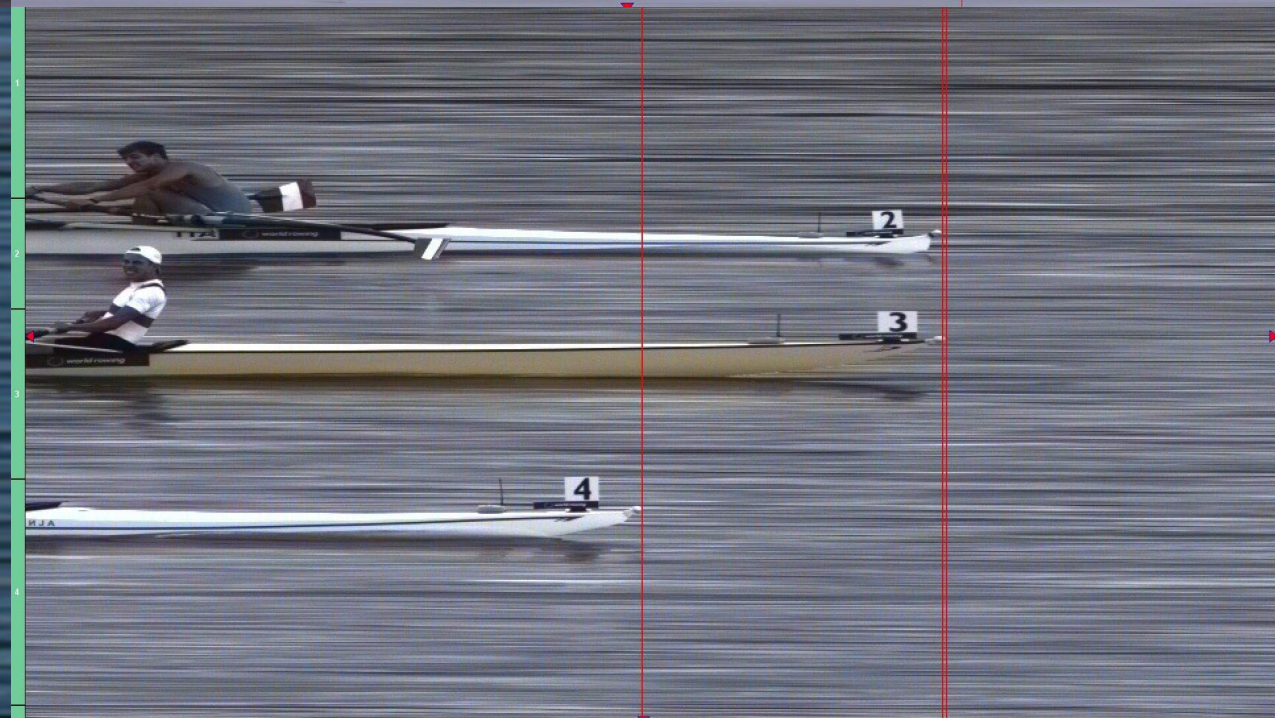
2016 Olympics

Omega sent 480 timekeepers and 450 tons of timing equipment to Rio

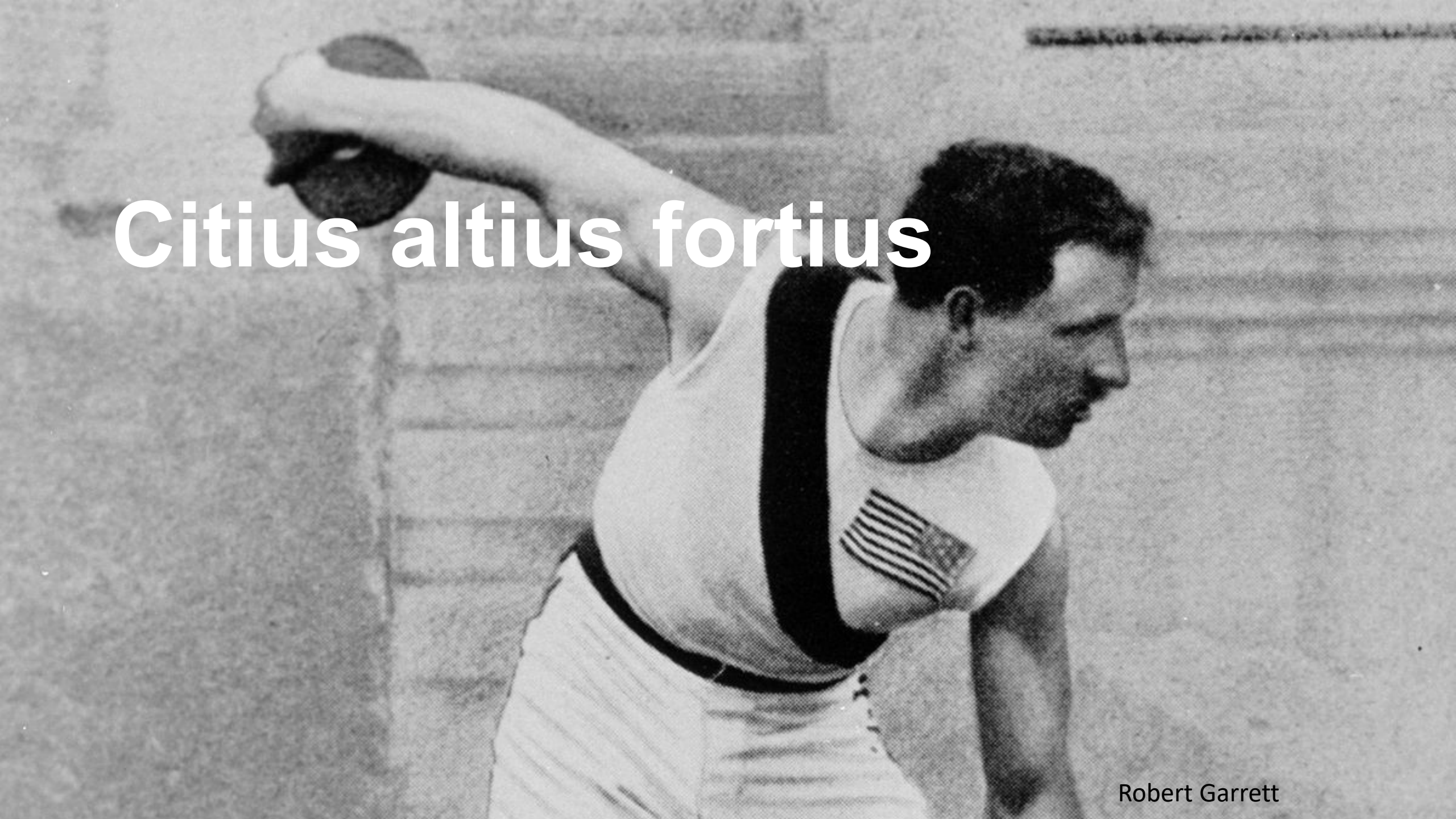
1
5
6


2
4
SWISS TIMING

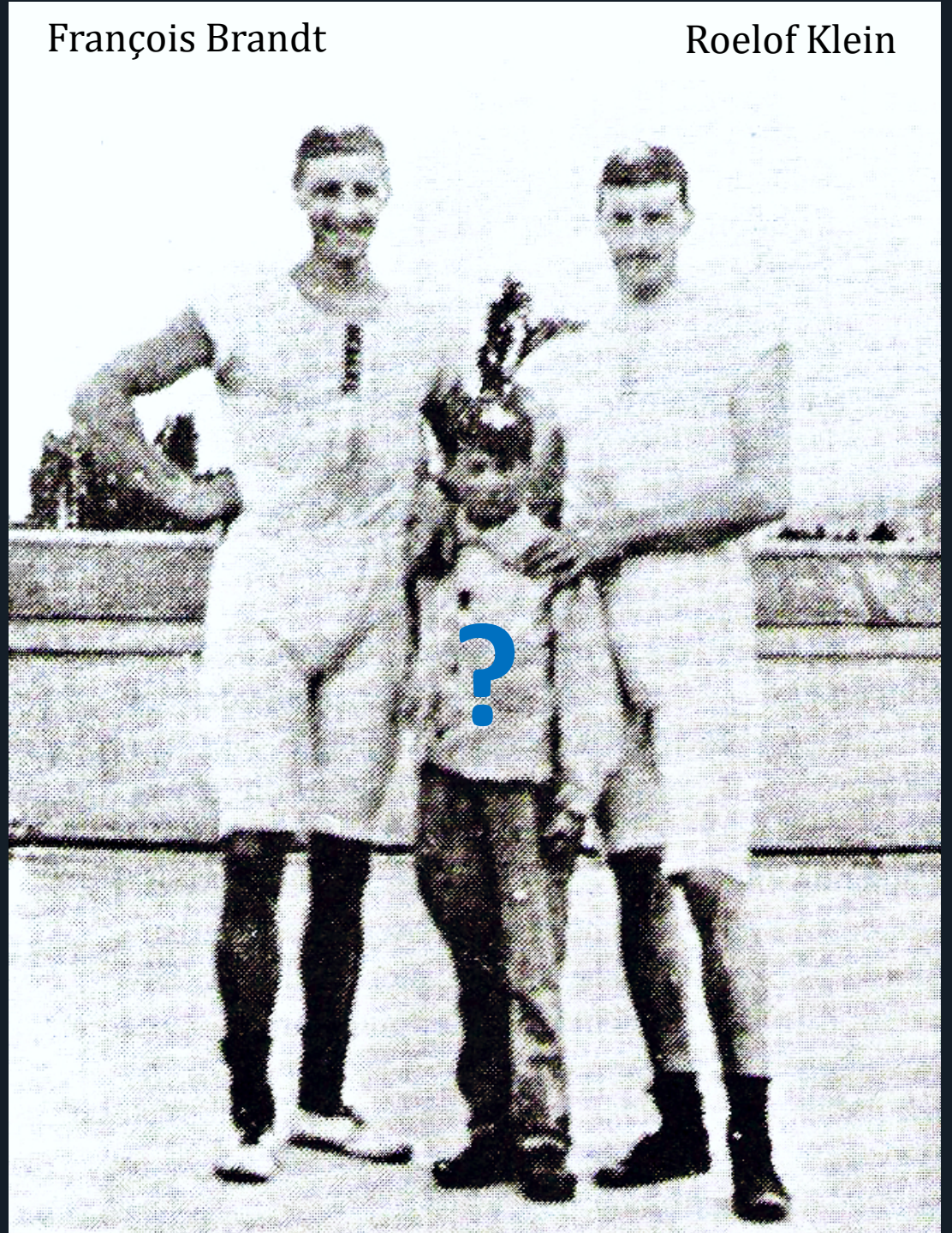
4
5
1 NZL M. DRYSDALE 6:41.34
2 CRO D. MARTIN 6:41.34
3 CZE O. SYNEK 6:44.10
OMEGA


2
3
4

Citius altius fortius



Robert Garrett



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Vladimir Kazantsev, USSR

Horace Ashenfelter- USA

Hard, Cold Numbers

Cold War

Televised sports-
+ soft, WARM sofas

Fast TV

Slow TV

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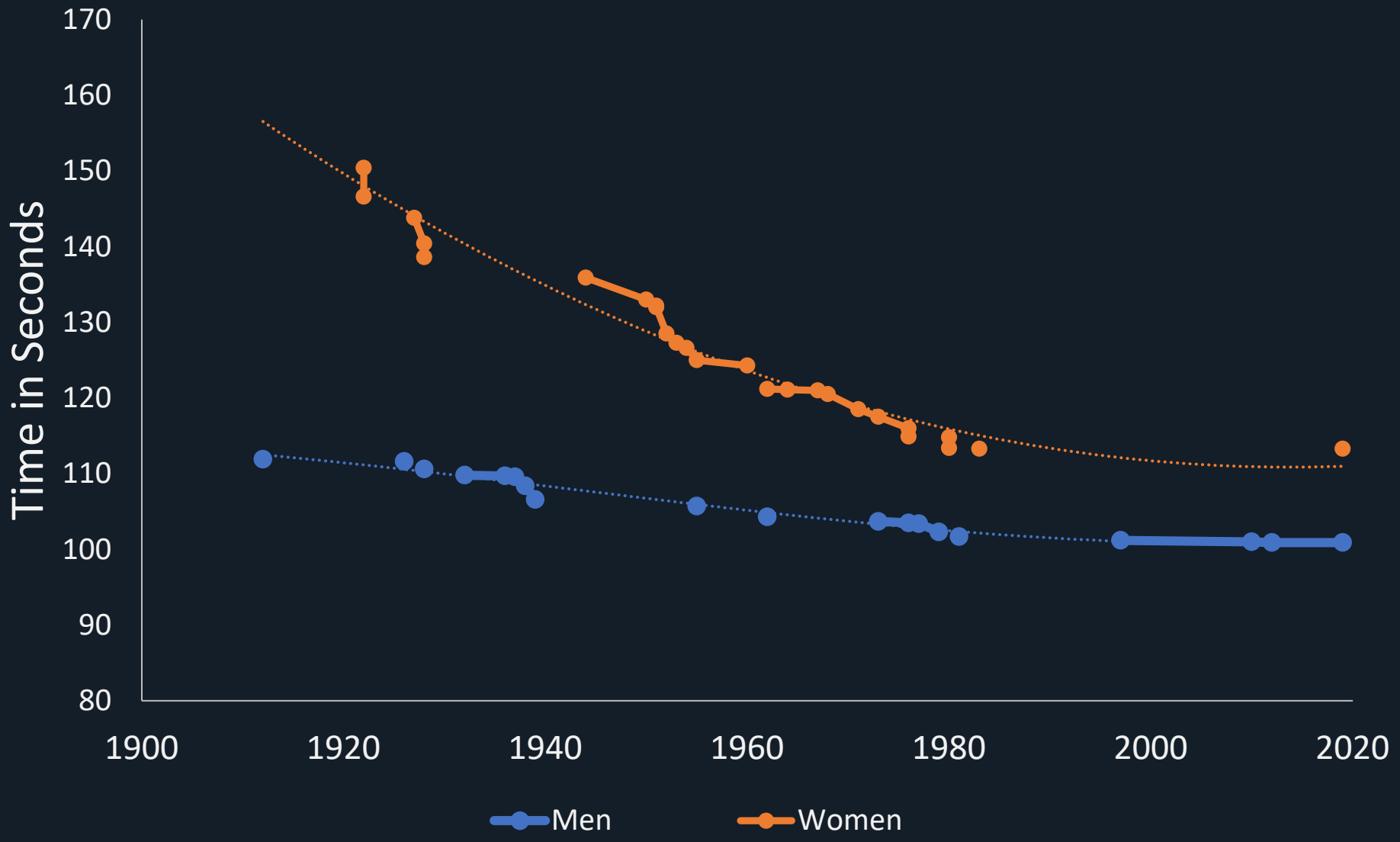
800meter WR Progression



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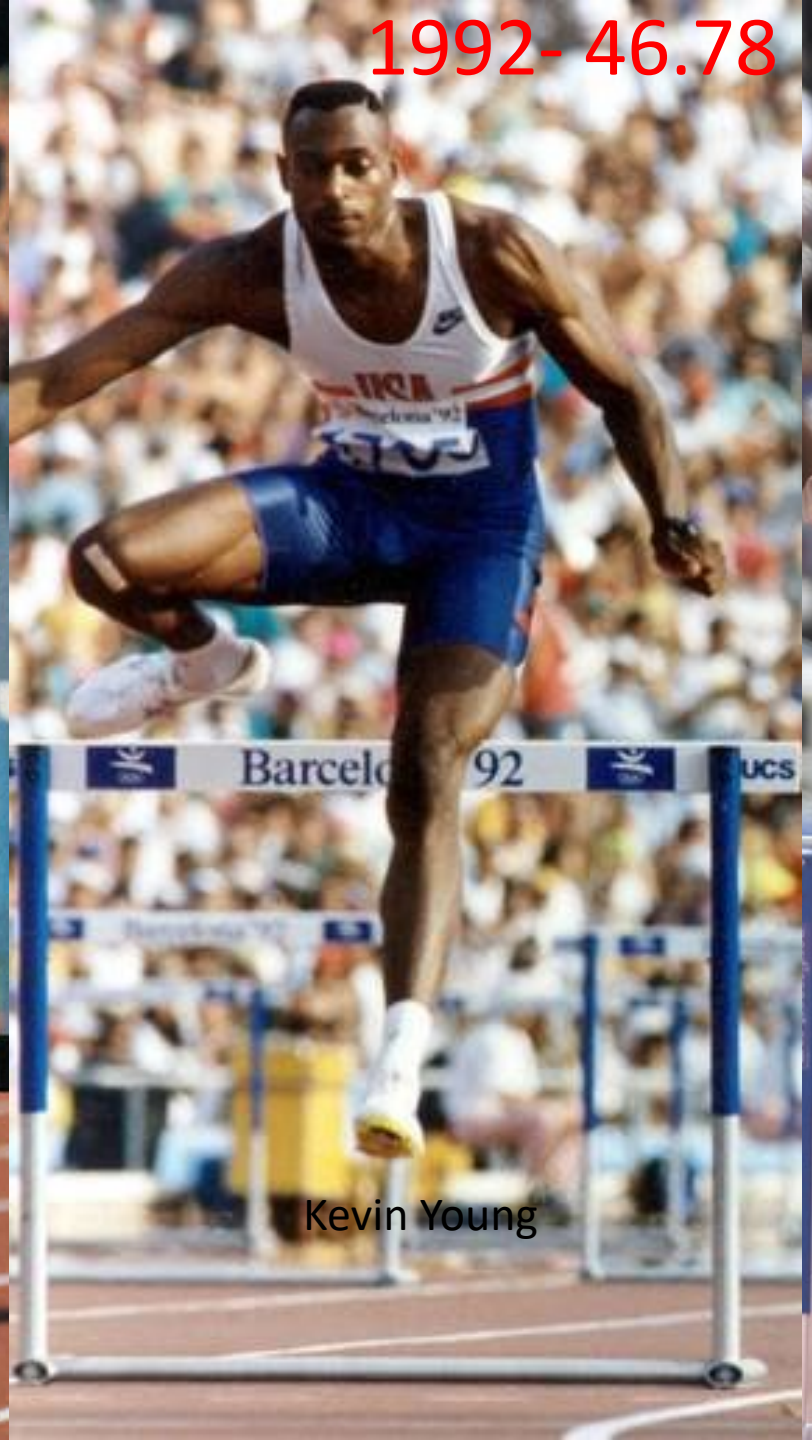
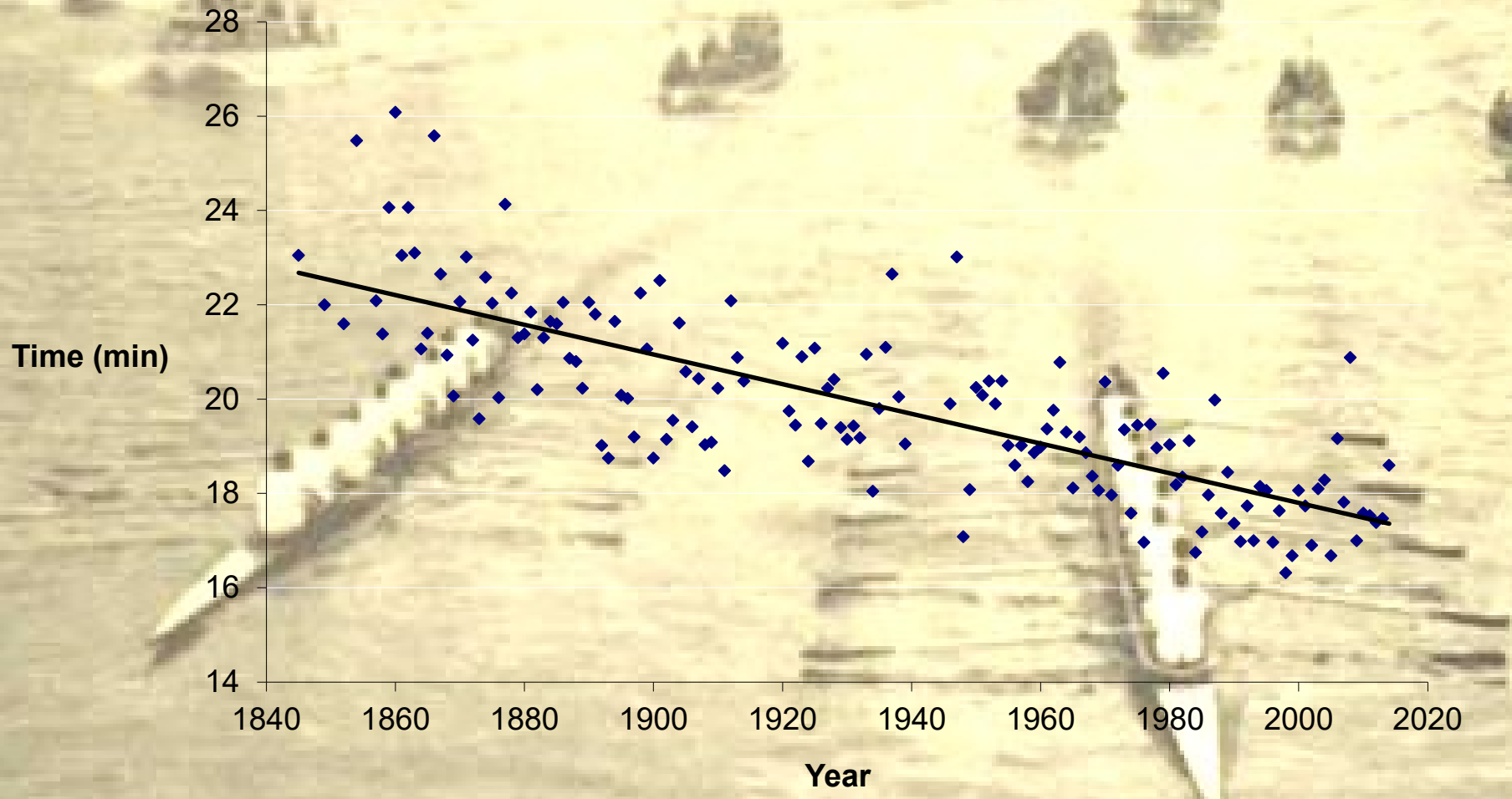
1983- 47.02
USA
KARHU
Edwin Moses


1992- 46.78
USA
Barcelo 92
Kevin Young

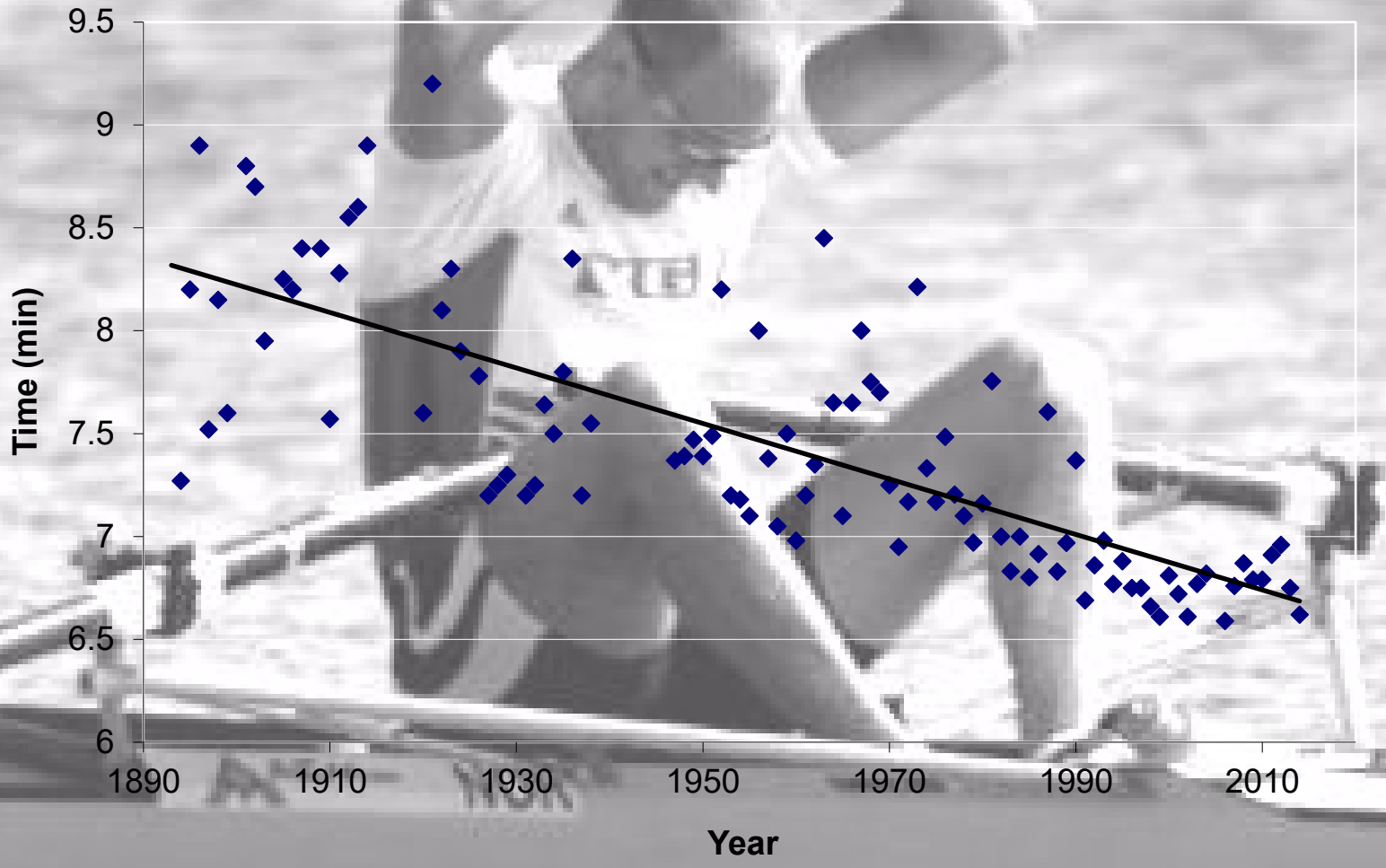
2019- 46.92
BAUHAUS
WARHOLM
BOSCH
BAUHAUS GALA
Karsten Warholm

Oxford-Cambridge Boat Race

Winning Times 1845-2014



FISA Men's championship 1x
Winning Times 1894-2014



TRAINING,

IN

THEORY AND PRACTICE.

BY

ARCHIBALD MACLAREN.

London:
MACMILLAN AND CO.
1866.

Rise at 7 a.m: Run 100-200 yards as fast as possible

	Tea, two cups, or towards the end of training a cup and a half only. Watercresses occasionally.	
Exercise (forenoon).	None.	
Dinner about 2 p.m.	Meat, Beef or Mutton. Bread.	
	Vegetables — Potatoes, Greens.	Some Colleges have baked Apples, or Jellies, or Rice-

About 5:30: Start for the river and row for the starting post and back

	Watercresses. Beer, one pint.	
Bed at 10.0		

Reckoning a half an hour in rowing to and half an hour from the starting point, and a quarter of an hour for the morning run- in all, say, one and a quarter hours.

Buzzword: *Innovation*

"Staying Relevant"

"Creativity is thinking of something new. Innovation is the implementation of something new"

"Fresh thinking that creates value"

"Innovation is significant positive change"

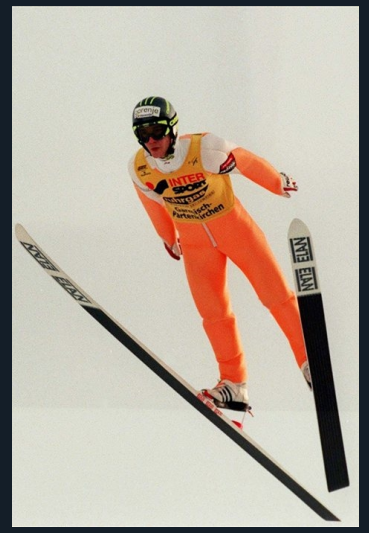
"Change that creates a new dimension of performance"

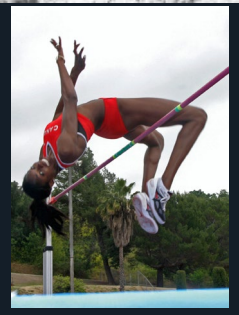
"The Future Delivered"

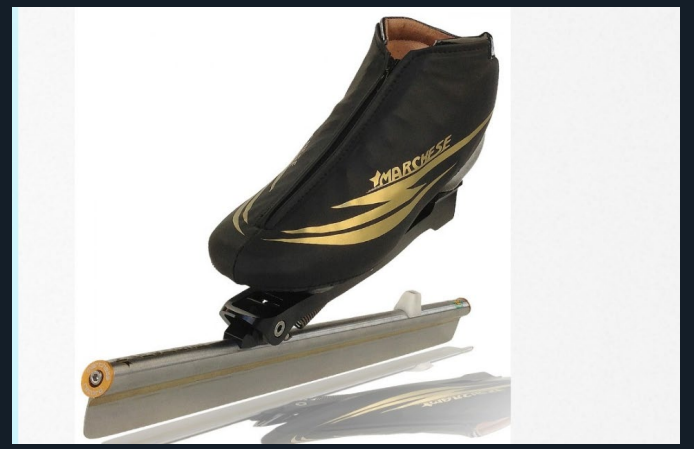
"Going from idea to value"

"Research is converting money to ideas. **Innovation** is converting ideas to money"

UiA


INTER
SPORT
ATOMIC
ATOMIC




MARCHESE

W. N. BLAKEMAN, Jr.

SLIDING ROW-LOCKS, &c.

No. 184,031. Patented Nov. 7, 1876.

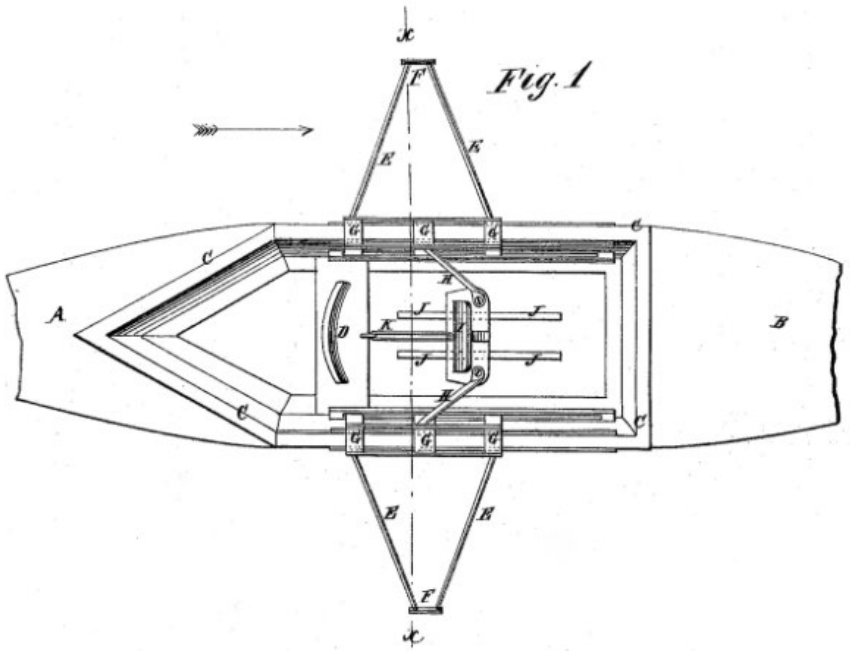
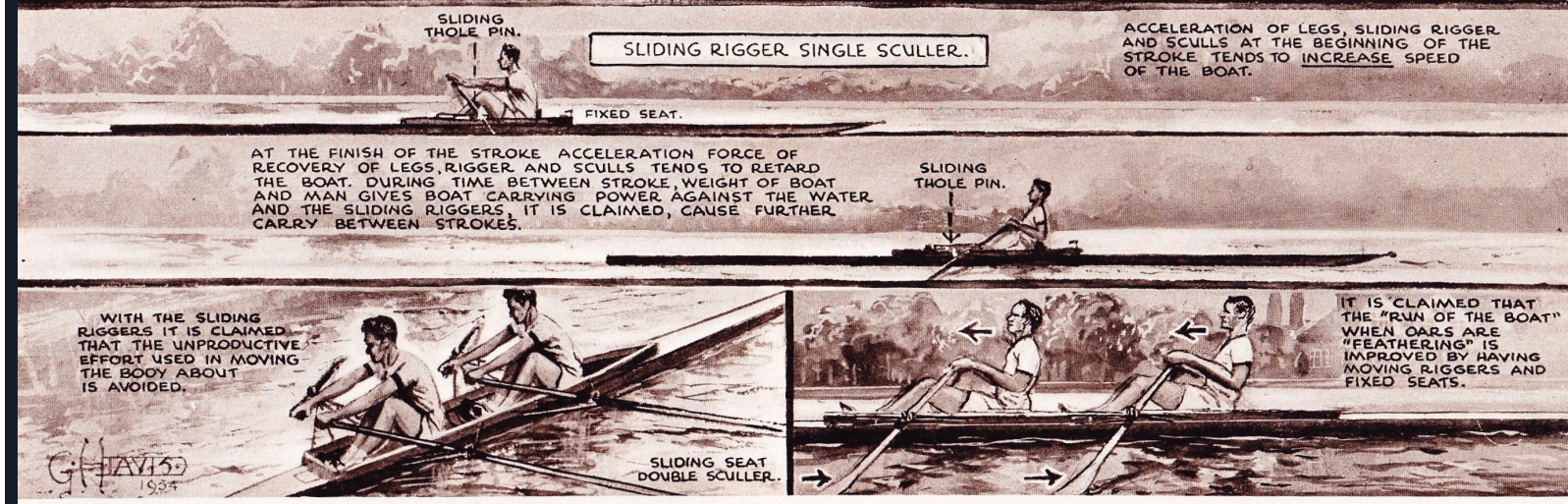


Fig. 1



ONE OF THE MOST INTERESTING FEATURES OF THE 1954 REGATTA SEASON; AND THE FIRST MAJOR ALTERATION IN BOAT DESIGN SINCE 1872: A SLIDING RIGGER INSTEAD OF A SLIDING SEAT—AN INVENTION WHICH MAY REVOLUTIONISE ROWING.

One of the most interesting features of this year's Regatta season, apart from some very excellent racing, has been a new invention, devised by Mr. C. E. Poynter, of Bedford Rowing Club, which may well revolutionise rowing. He has fitted a seat in a double-sculler which, instead of sliding, is fixed, while the rowlocks are attached to a sliding cradle which also incorporates the stretcher. The whole of this cradle slides up and down the boat on metal runners. And so, instead of the sculler moving backwards and forwards on his sliding seat, it is the stretcher and riggers which move. By this means it is claimed that the boat is given more momentum. One of the drawbacks of the sliding seat, which the fixed seat will obviate, is that the crew, which weighs considerably more than the boat, has to move some 18 ins. backwards and forwards every stroke, thus shifting the centre of buoyancy of the boat. This causes the bows and stern to dip alternately some two or three inches into the water each stroke. With the sliding rigger the crew no longer has to shift its own weight about unproductively, the momentum of the boat is almost unimpaired, and the crew saves a lot of energy. The sliding seat was introduced in 1872, and since then no major improvement in the design of rowing and sculling racing boats has been made. It must have been a disappointment to all oarsmen that Mr. Poynter could not get his boat ready for this year's Henley Regatta. It remains to be seen, therefore, whether his sliding rigger will prove efficacious when on trial under racing conditions.

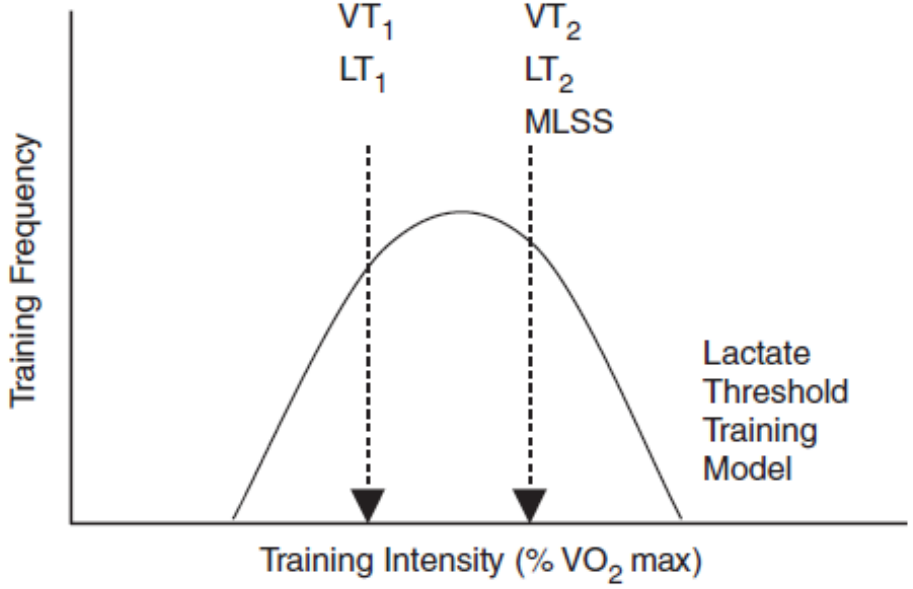
DRAWN BY OUR SPECIAL ARTIST, G. H. DAVIS.

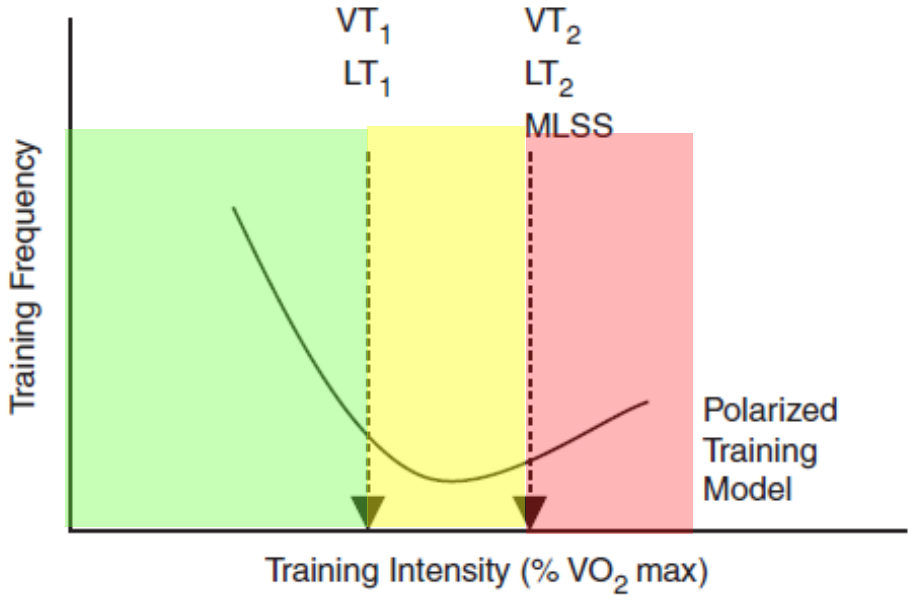
UiA

Source for all 3 images: https://heartheboatsing.com/2017/10/17/geek-special-the-ups-and-downs-of-the-sliding-rigger/

80:20 Endurance Training

Polarized Training





Seiler & Kjerland. *Scand. J. Med. Sci. Sports*. 16, 49-56, 2006.

Photo courtesy of Dr. Dajo Sanders & Team Sunweb

Sports Technology Innovation

Solving a **specific and real problem** experienced by athletes or coaches and **improving** the **sports performance/development** process in a measurable way

New technologies should pass this test:

This new technology solves
THIS problem for THESE Athletes
BETTER than what we did before

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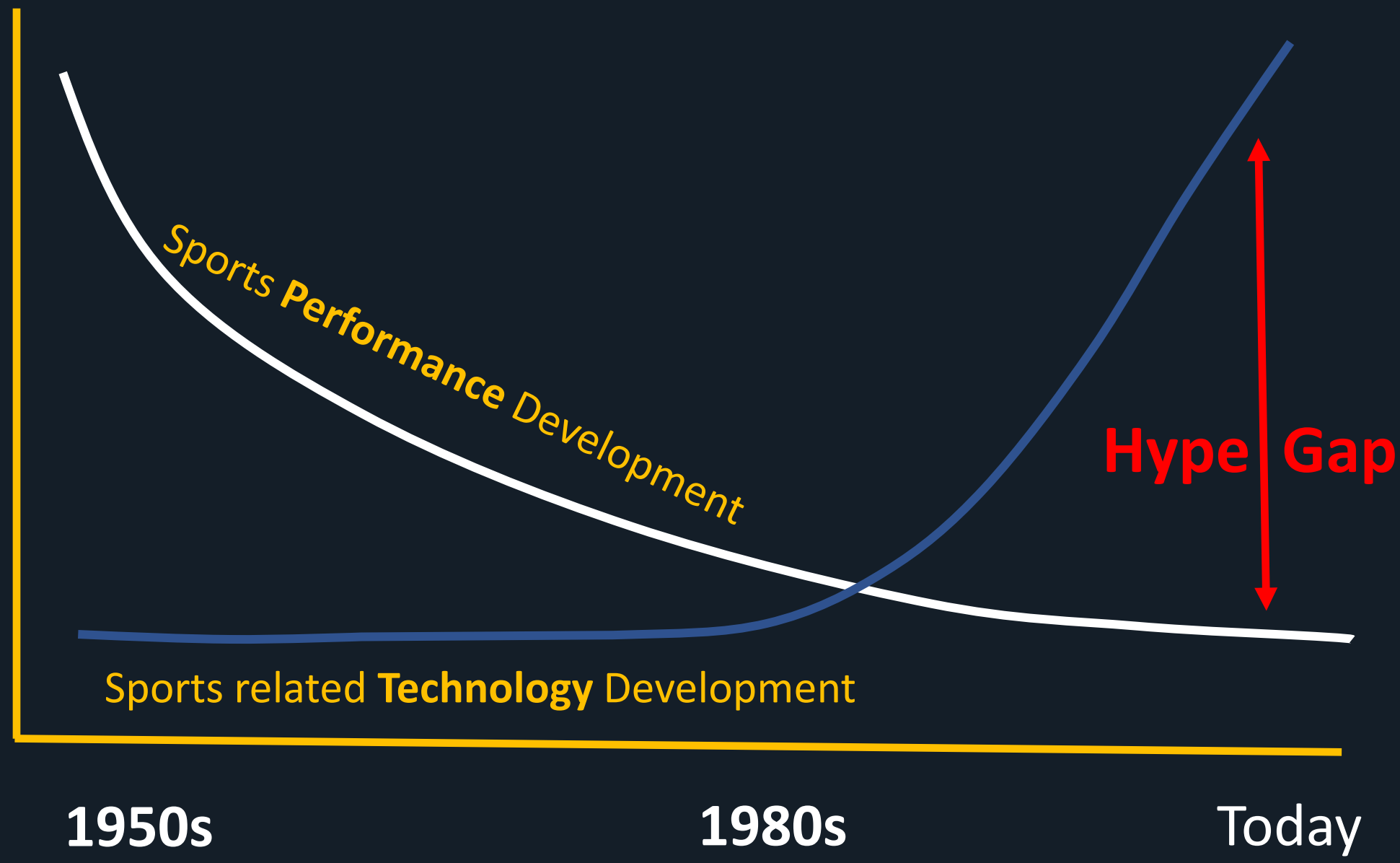
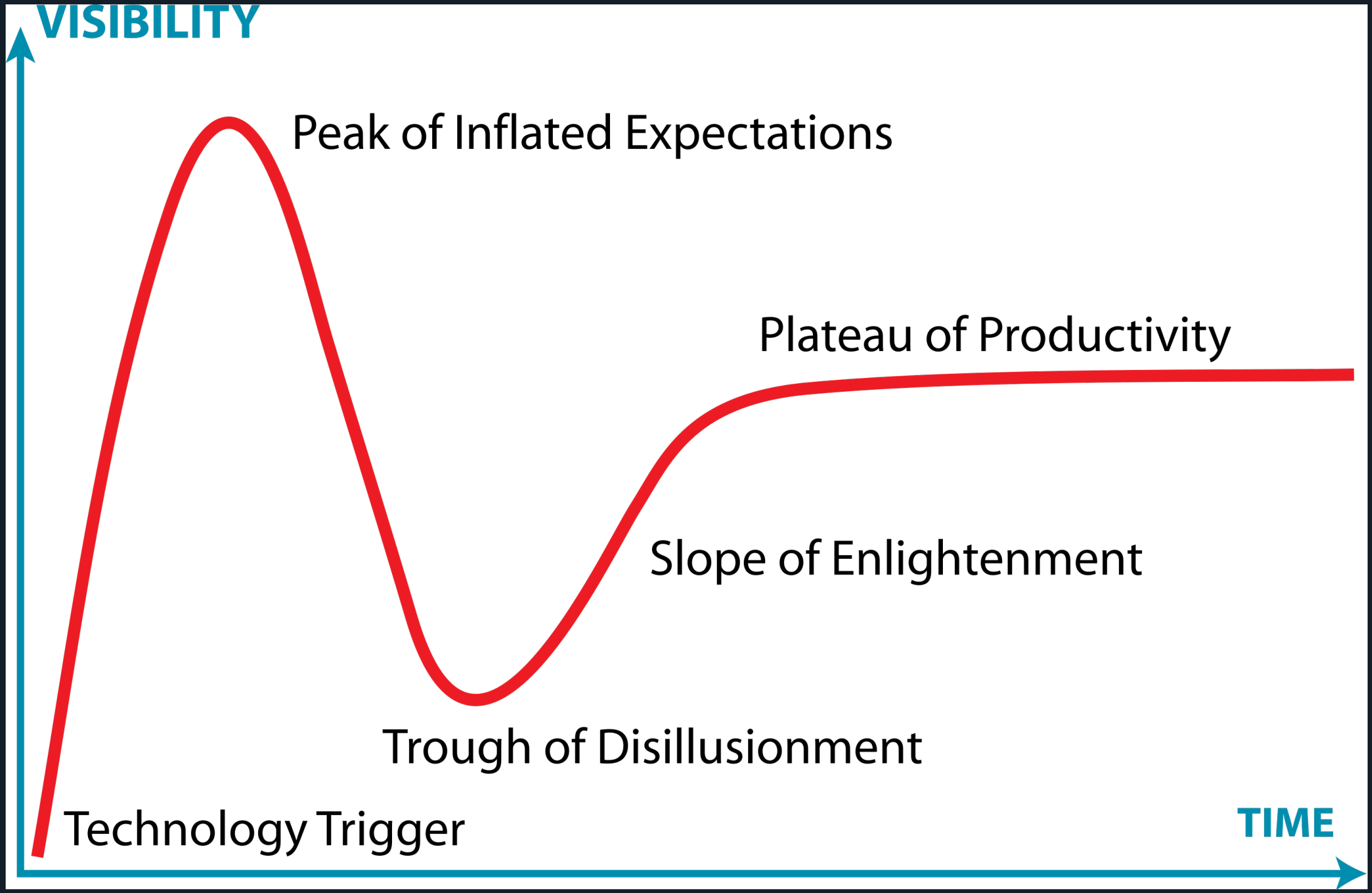

Sports Performance Development
Hype Gap
Sports related Technology Development
1950s
1980s
Today

UiA


VISIBILITY
Peak of Inflated Expectations
Plateau of Productivity
Slope of Enlightenment
Trough of Disillusionment
Technology Trigger
TIME

UiA

Apollo 13

"Houston,
we've had a problem"

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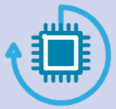
Current technological development directions impacting sports science and training

Home and field based physiological measurements

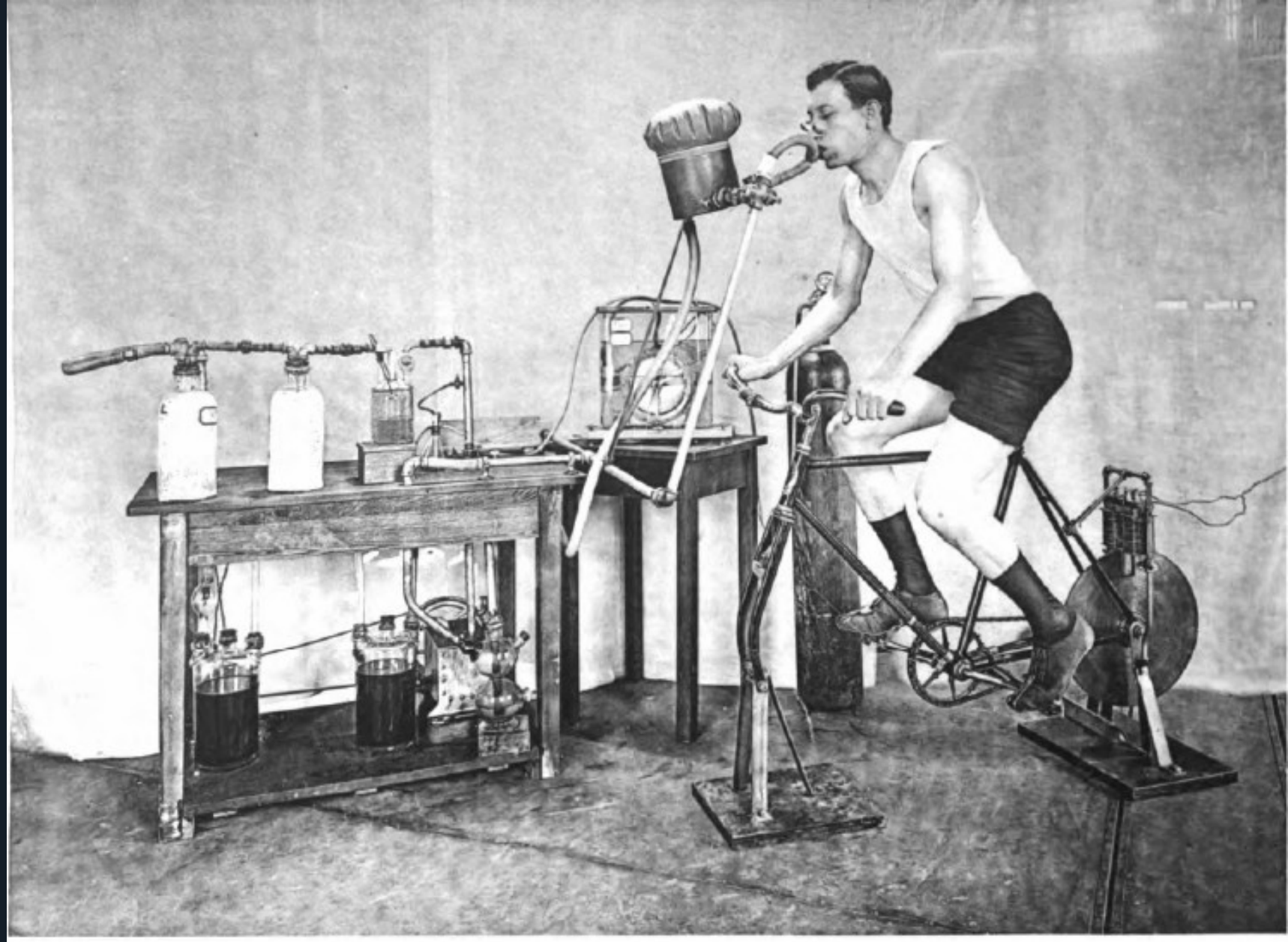


Embedded movement sensors (limbs, rackets, bats, boats, teams etc.)

“Real-time” movement analysis

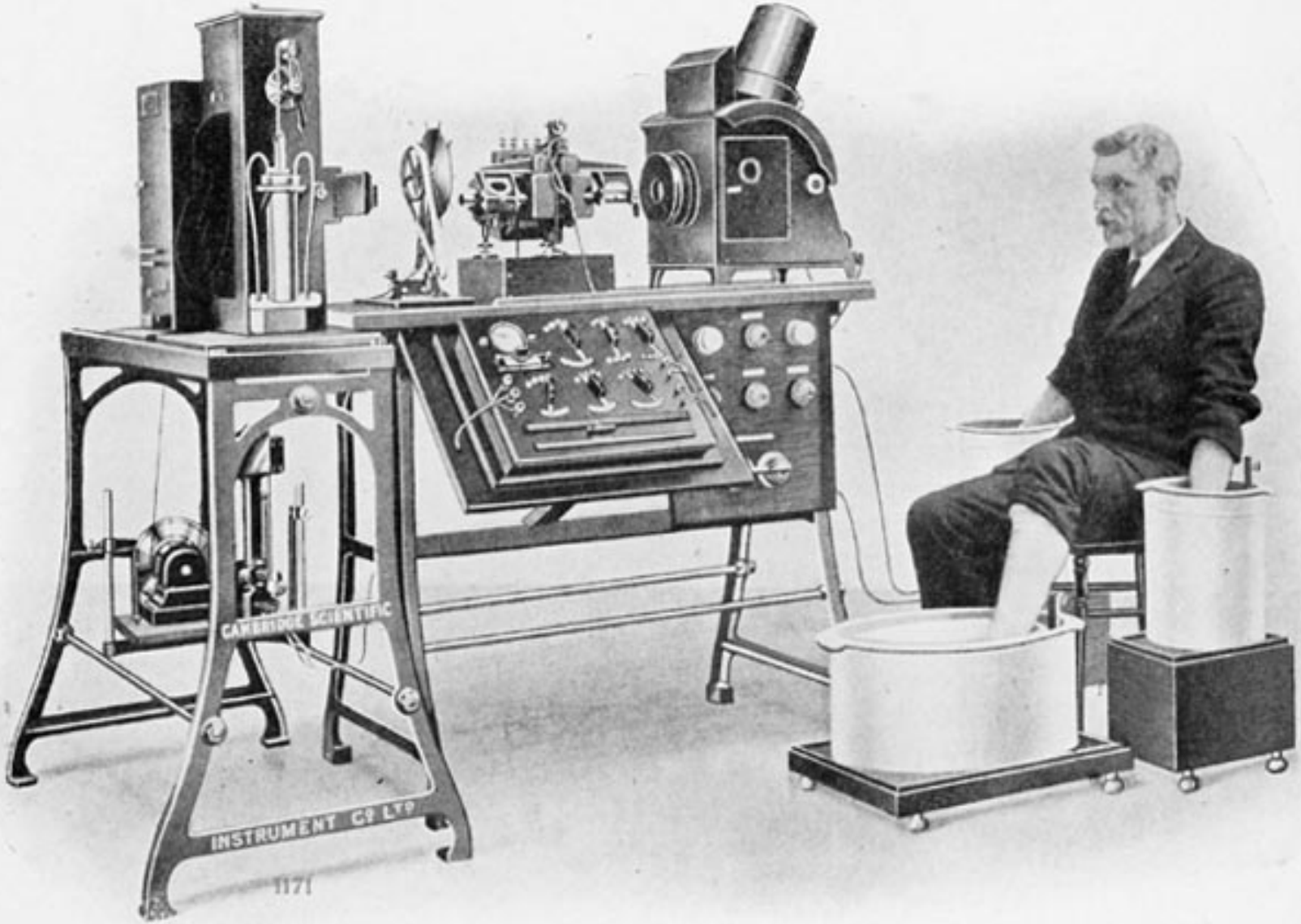
Big (Training & Performance) Data analytics-
hypothesis driven and “hypothesis free” approaches

Cellular/molecular interventions- to accelerate recovery or enhance adaptations



BICYCLE ERGOMETER AND UNIVERSAL RESPIRATION APPARATUS.

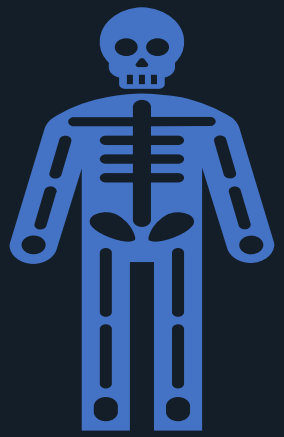
The subject riding bicycle ergometer with an electric brake breathes through a mouthpiece into a current of air passing through a pipe system. The expired air is forced by a rotary blower through sulphuric acid and soda lime to remove respectively carbon dioxide and water. The carbon-dioxide absorbers may be weighed. Measured amounts of oxygen from a cylinder are passed through an accurate meter and introduced as needed.

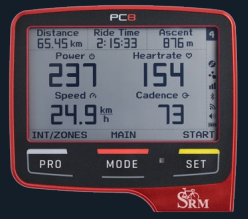


PHOTOGRAPH OF A COMPLETE ELECTROCARDIOGRAPH, SHOWING THE MANNER IN WHICH THE ELECTRODES ARE ATTACHED TO THE PATIENT, IN THIS CASE THE HANDS AND ONE FOOT BEING IMMERSED IN JARS OF SALT SOLUTION

Ruldolf Harbig
WR 800m 146.6
1940-1955

Physiology in the Field





Power/Pace

Physiological Responses

External Work

Internal "Cost"

Perceived Effort/ Exertion

6	No exertion
7	
8	
9	
10	
11	Light
12	
13	Somewhat hard
14	
15	Hard (heavy)
16	
17	Very hard
18	
19	
20	Maximal exertion

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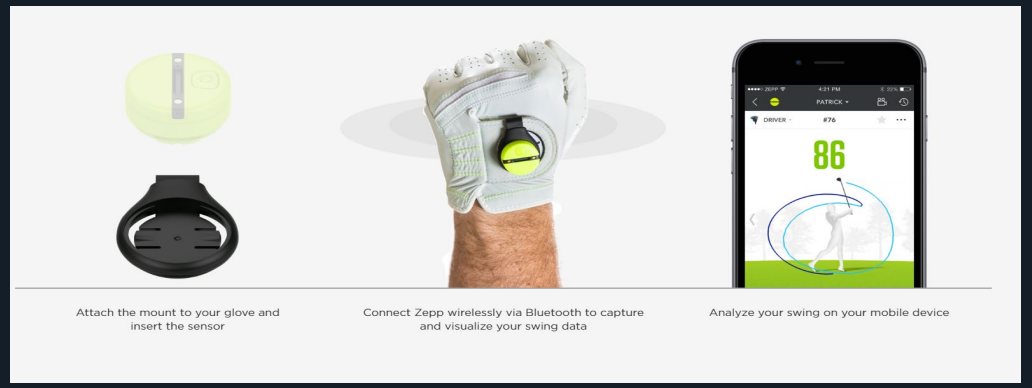
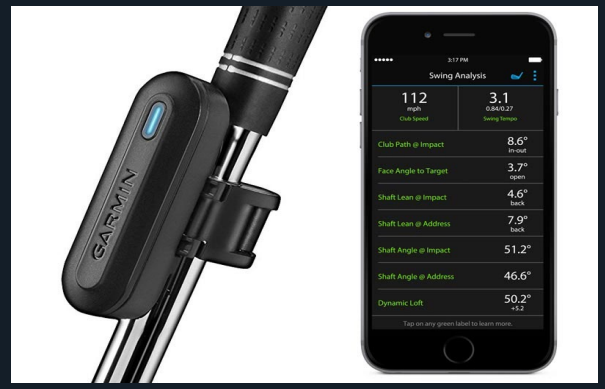
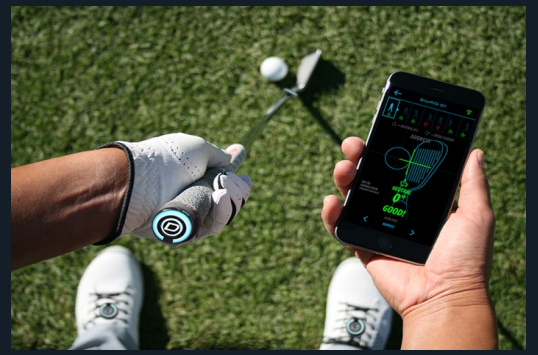
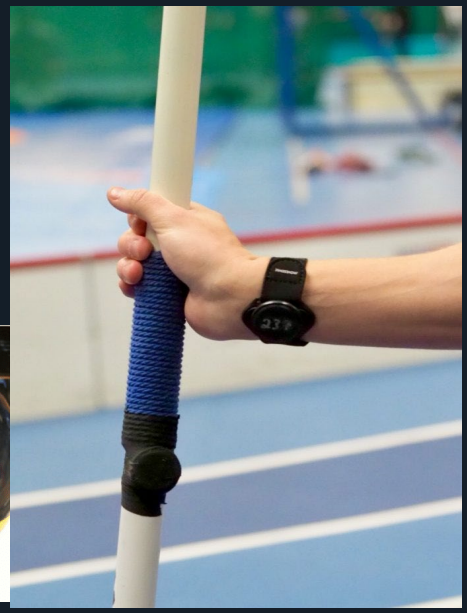
ZWIFT
Stephen Seiler
CHANGE VIEW
ZWIFT
NEOBIKE
Uno X
ACTIVE
NEOBIKE

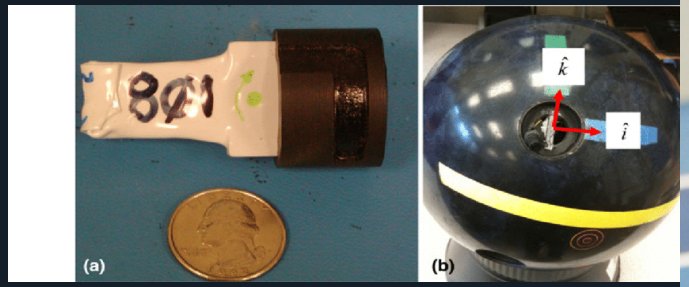
Here a sensor, there a sensor, everywhere a sensor





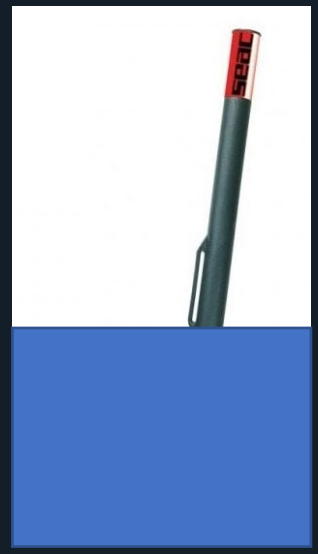






Early 1990s: My **rowing**
Boat Velocity Fluctuation Sensor:

(Speed Fountain™)



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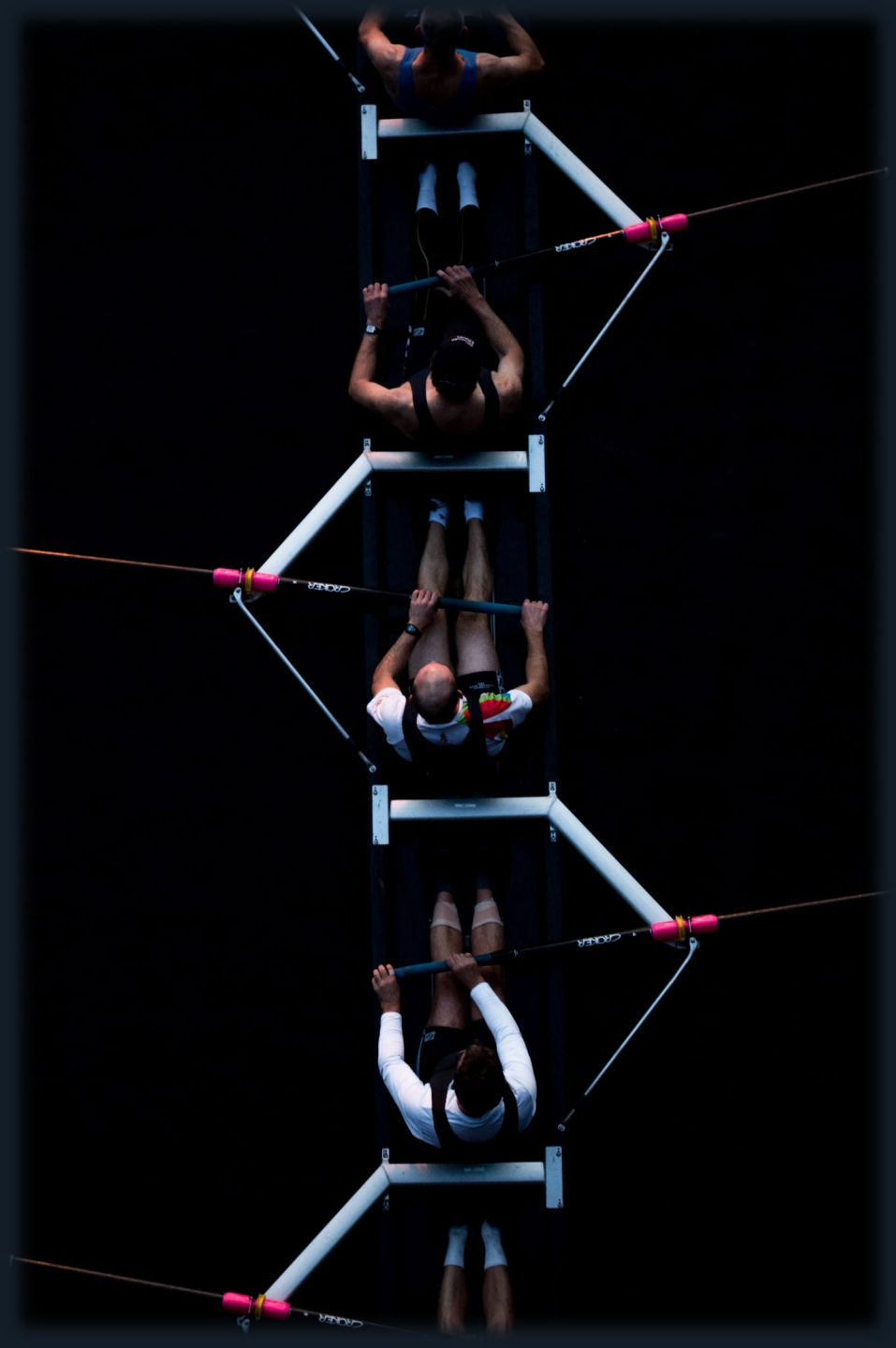
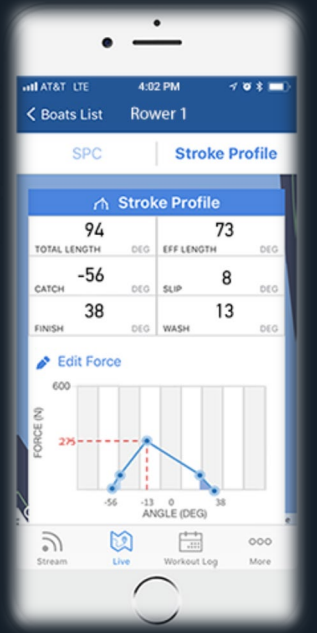
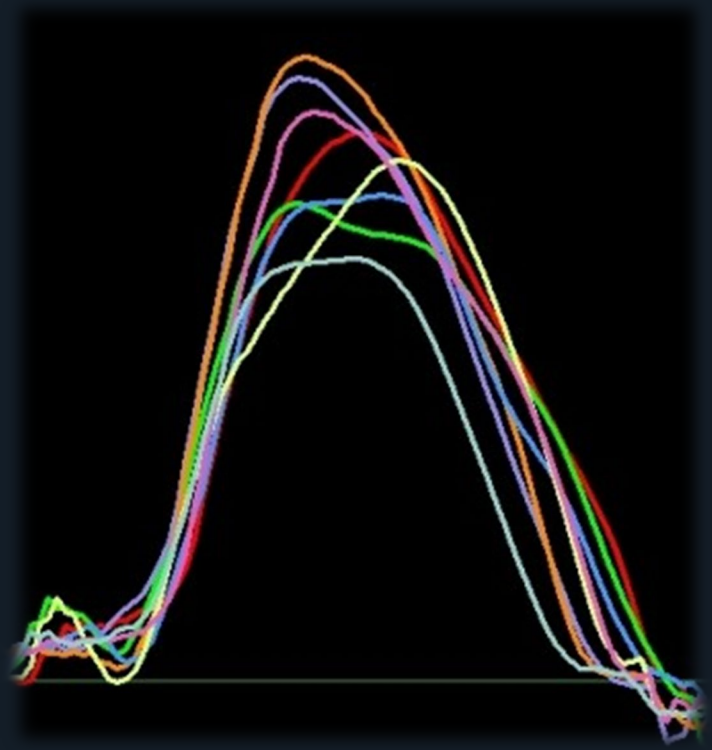
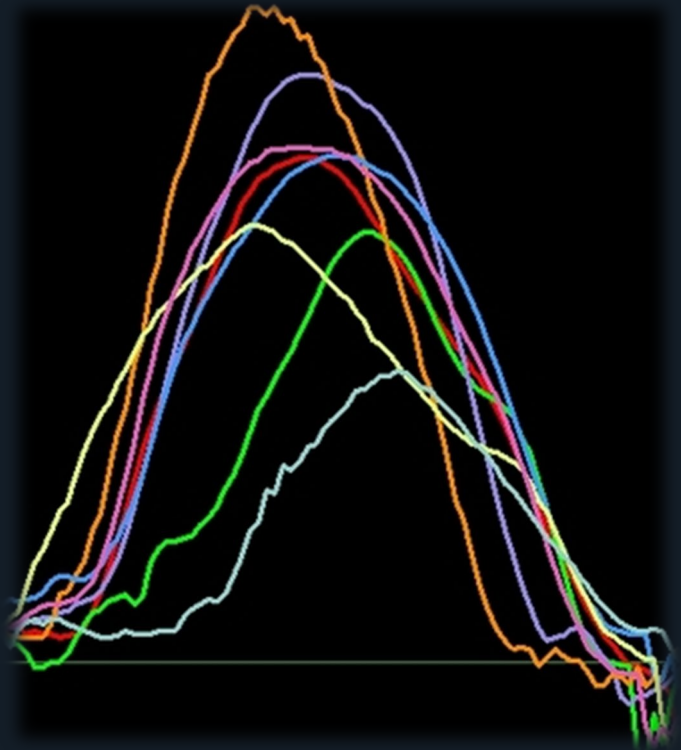
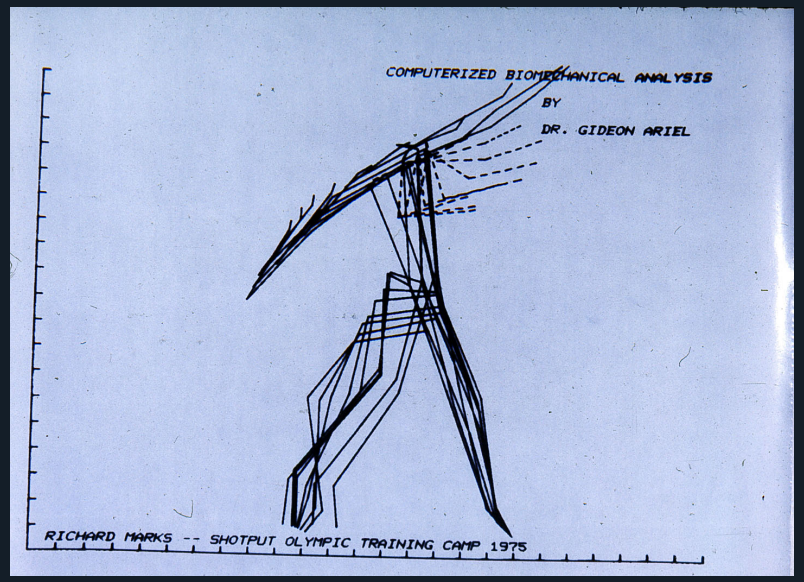



AT&T LTE
4:02 PM
Boats List
Rower 1
SPC
Stroke Profile
Stroke Profile
94
TOTAL LENGTH
DEG
73
EFF LENGTH
DEG
-56
CATCH
DEG
8
SLIP
DEG
38
FINISH
DEG
13
WASH
DEG
Edit Force
600
FORCE (N)
275
-56
-13
0
38
ANGLE (DEG)
Stream
Live
Workout Log
More

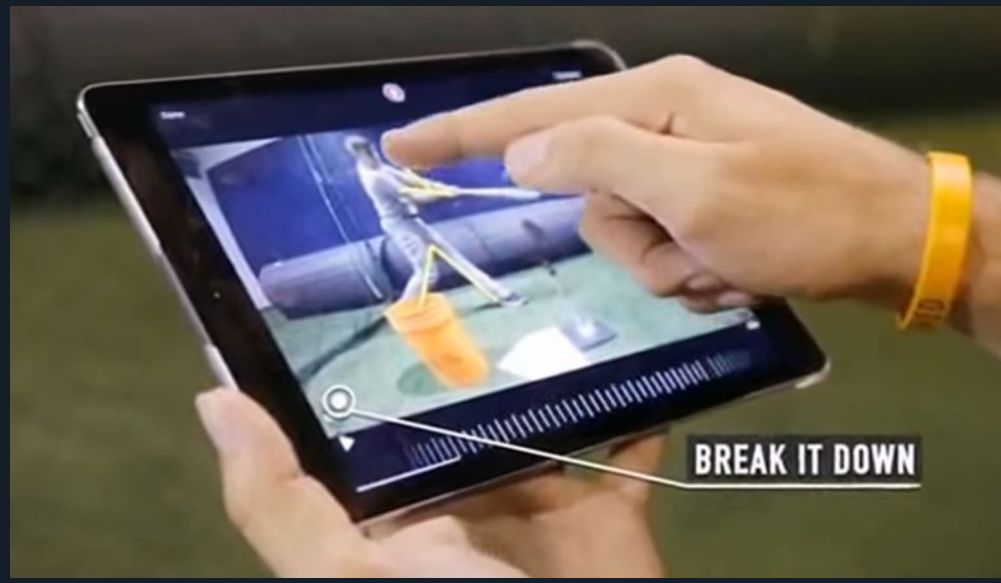




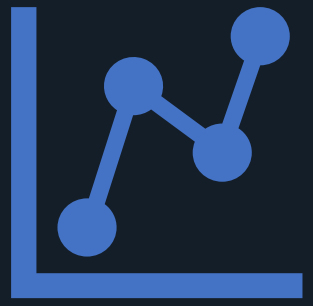
1970s-1980s



TODAY



UiA



From almost no data to
data overload
in 25 years

Rankings

Calculate | Download

Download Type:
(•) Long ( ) Wide

Ergometer Type:
Rower

Group data by:
Age Band

Sub-group data by:
Event

Gender:
Male

Weight:
All

Event:
500m 2000m

Min Matches:
1 — 4 — 9

Filter By:
None

Filter Value:

Season:
2002 — 2019

Top Percent:
0 — 94 — 99

Power Range:
0 — 1400

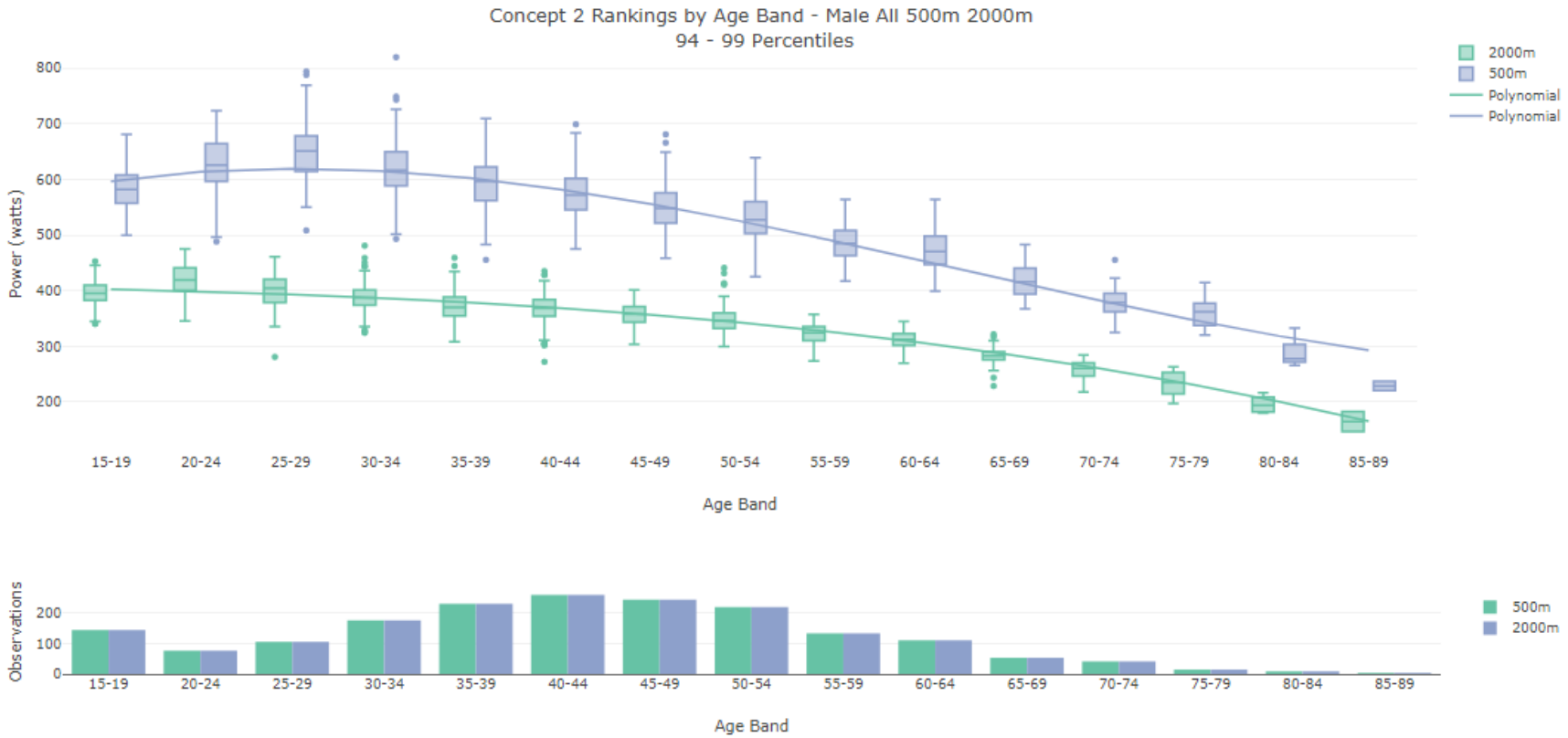
Concept 2 Rankings | Notes

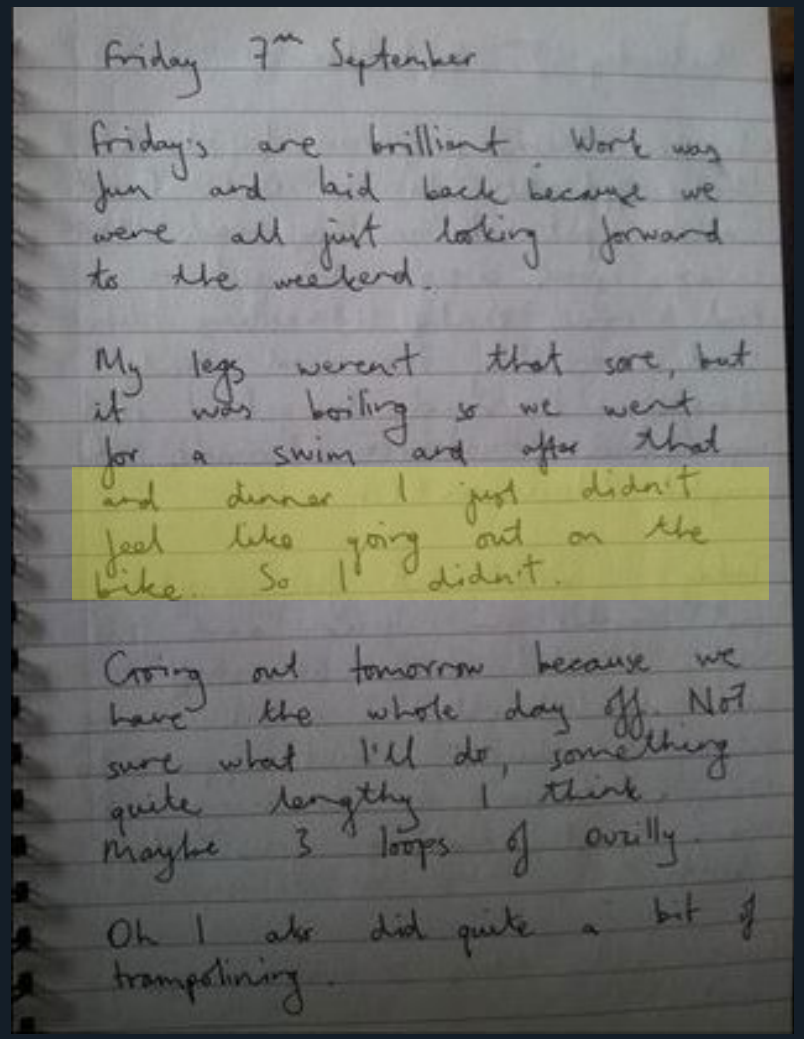
Log out

Data provided by the log.concept2.com website as of 09/05/2019 - Don't forget to click the 'Calculate' button

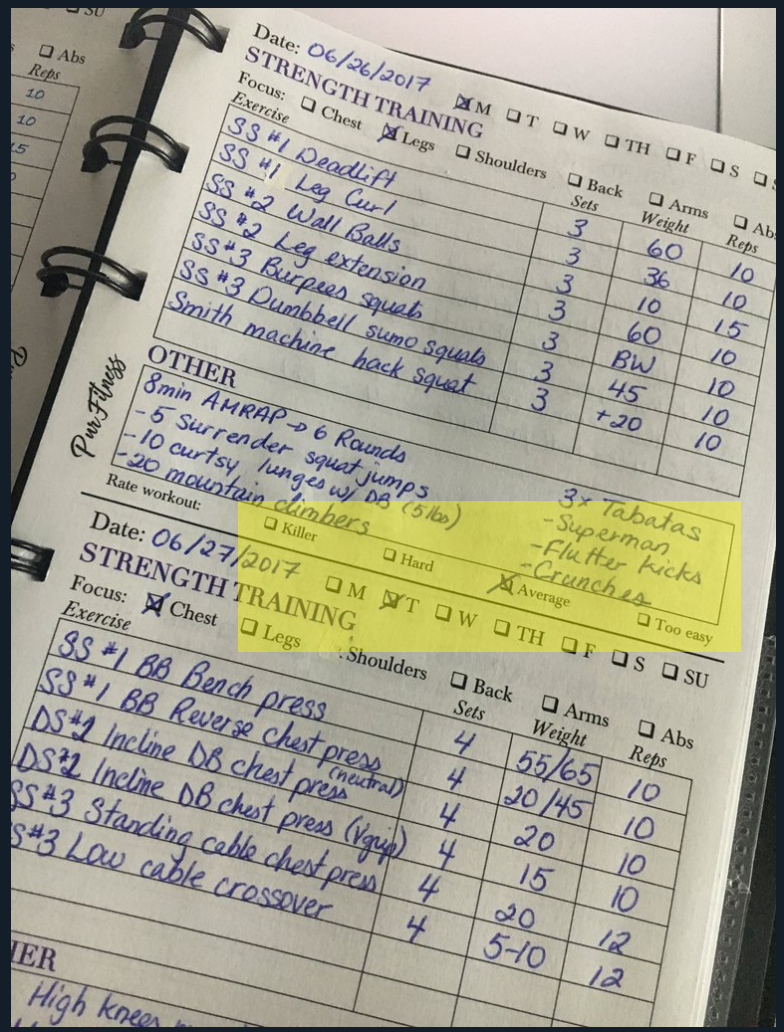
Chart Type:
(•) Box Plot ( ) Scatter ( ) Bar ( ) Percentile lines

Measure:
(•) Power ( ) Time ( ) Distance ( ) Split ( ) Age ( ) Experience





Diary of Training

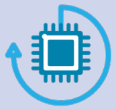


Logbook

Digital Metrics Algorithms

How do you feel?
How are you responding to the training?

Home and field based physiological measurements



Embedded movement sensors (limbs, rackets, bats, boats, teams etc.)

“Real-time” movement analysis

Big (Training & Performance) Data analytics-
hypothesis driven and “hypothesis free” approaches

UiA

Feed back

UiA

NORGE

Olympiatoppen

Competition
Execution

Relationships
in Performance Team

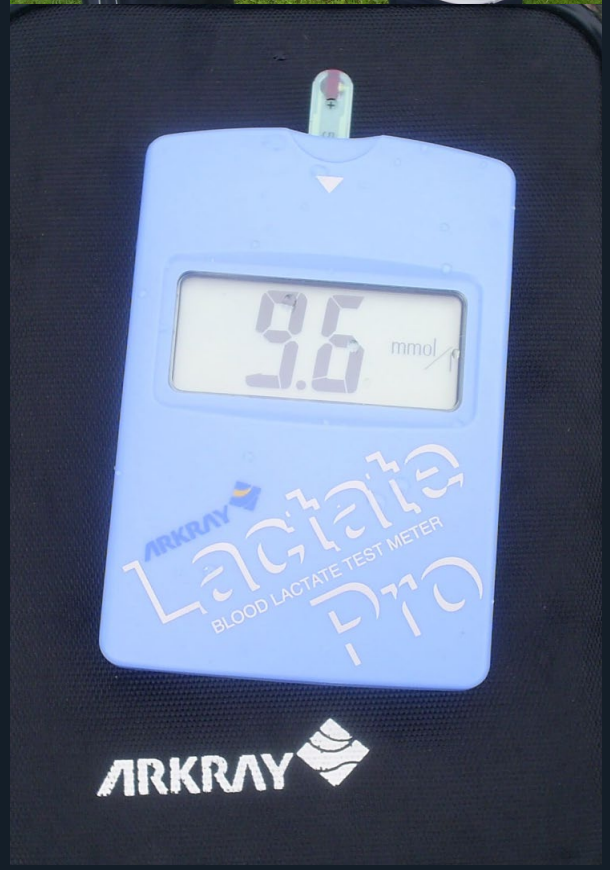
Daily Training
Quality

UiA

Does the feedback technology improve training quality by informing the communication between coach and athlete?

UiA

Telfort.
AEGON


9.6
mmol/l
ARKRAY
Lactate Pro
BLOOD LACTATE TEST METER
ARKRAY

Good training sessions happen when **External** and **Internal** feedback align!

The coach said it

I felt it

The video camera showed it

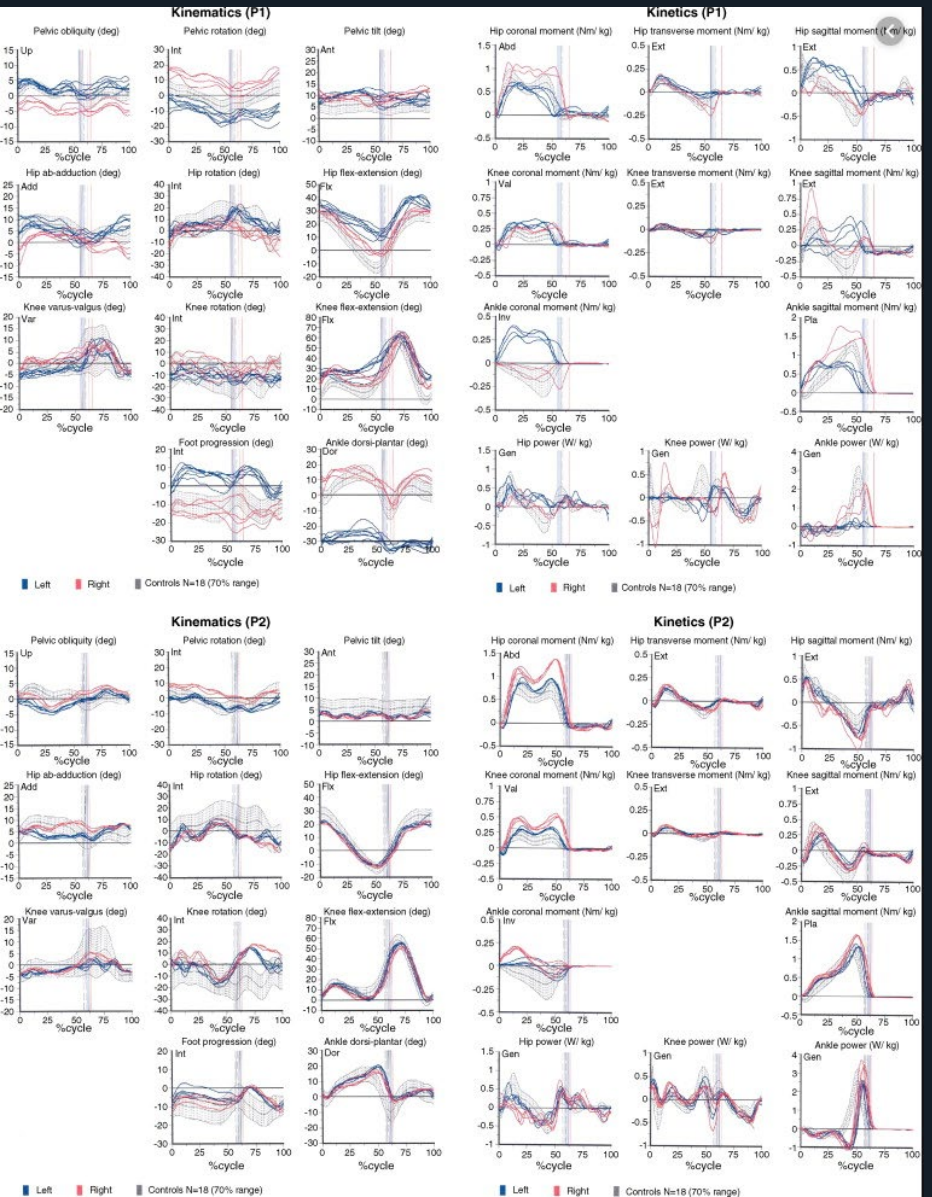
Source: Siren Sundby, 2004 Olympic Gold medal winner and Arne Riise, Olympiatoppen

We have access to numerous technologies that can **potentially** improve the overall quality of daily training. If they do not, then they are not useful.

This is the test that each technology must pass!

Sports Scientist

Athlete



When the competition begins.....

they have to depend on what they feel

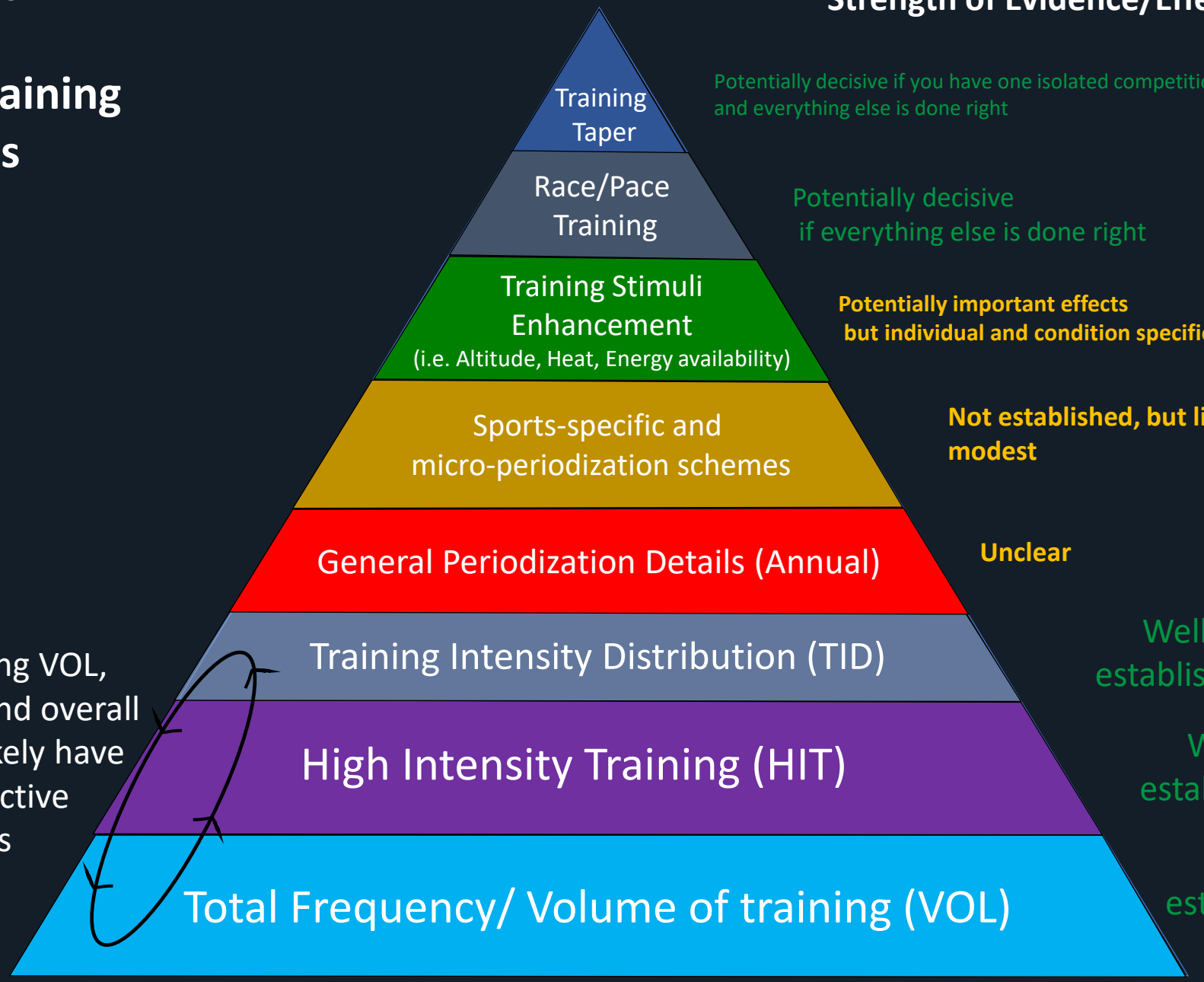
Cellular & Molecular «Technologies»

Accelerate Recovery

Enhance Training Stimulus

UiA

Hierarchy of Endurance Training Priorities



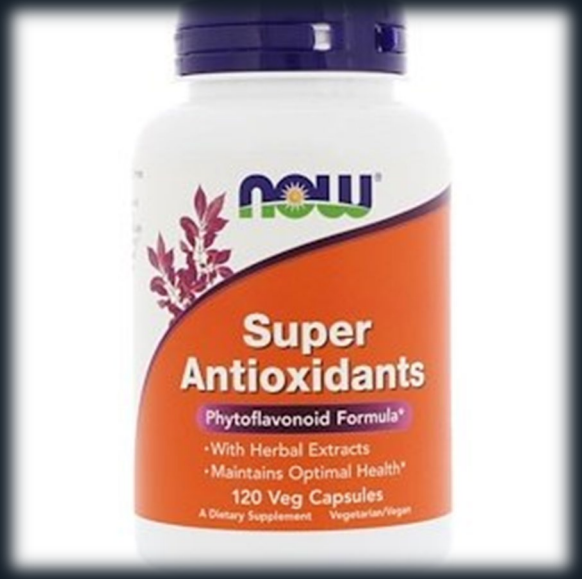
UiA

↑Core Temperature– heat stress

↓Gycogen - energy stress

↑Reactive Oxygen Species (ROS)

↑Inflammation Response


now
Super Antioxidants
Phytoflavonoid Formula*
•With Herbal Extracts
•Maintains Optimal Health*
120 Veg Capsules
A Dietary Supplement
Vegetarian/Vegan

TROPICAL FRUIT
CONCORD GRAPE
APPLE
STRAWBERRY
DRAGONFRUIT
GOJI
BLUEBERRY
CITRUS
WILDBERRY
POMEGRANATE
LG
ENERGY
ENDURANCE
GEL
GARNEAU

Advil
Ibuprofen Tablets, 200 mg
Pain Reliever/Fever Reducer (NSAID)
100 Coated Tablets
3,000 Points
TWIN PACK
X 100=200
Well at Walgreens
WALGREENS PHARMACIST RECOMMENDED
Ibuprofen
Ibuprofen Tablets USP, 200 mg /
Pain Reliever / Fever Reducer (NSAID)

WASHINGTON REDSKINS
EQUIPMENT

Our biology is a complex result of millions of years of evolution.

Efforts to shortcut this **biology** with profit Driven biotechnology are often highly oversimplified failures!

Thank you for
listening!